CSc 8830: Computer Vision

Module 4 Assignment

Classical Human Boundary Segmentation (RGB & Thermal) vs. SAM2, and Fourier-Domain Edge/Region Theory

GitHub repository: https://github.com/adebiyi22/Computer-vision

1. Assignment

1. **Part 1:** Find the exact boundary of a human in an RGB image using classical OpenCV functions only (no ML/DL); compare against SAM2.
2. **Part 2:** Same, but for a thermal/infrared image; compare against SAM2.
3. **Theory:** Derive how edge detection and region segmentation can be achieved via Fourier-domain analysis.

2. Implementation

2.1 Part 1 — RGB Segmentation (Classical CV)

`rgb_segmentation.py` uses only classical, non-learned operations: bilateral filtering, automatic (median-based) Canny edge detection, morphological closing to connect the boundary, the largest contour as a rough person estimate, and GrabCut (graph-cut energy minimization fit per-image, not a trained model) to refine the final silhouette.

2.2 Part 2 — Thermal Segmentation (Classical CV)

`thermal_segmentation.py` exploits the fact that warm bodies appear bright in thermal imagery: Otsu's automatic thresholding (minimizes within-class intensity variance - a classical, non-learned method) separates foreground from background, followed by morphological cleanup and connected-component analysis to isolate the largest warm blob as the person.

2.3 SAM2 Comparison (external baseline)

`compare_with_sam2.py` runs Meta's SAM2 model (a deep-learning model, used ONLY as an external comparison baseline, not as part of this assignment's own implementation) on the same image with a point prompt, and computes Intersection-over-Union (IoU) and the Dice coefficient between the classical mask and SAM2's mask.

3. Theory: Edge Detection and Region Segmentation via the Fourier Domain

3.1 The Differentiation Theorem

Differentiating f(x,y) under the inverse Fourier transform integral gives $F\{df/dx\} = (j2\pi u)F(u,v)$ and $F\{df/dy\} = (j2\pi v)F(u,v)$. Differentiation corresponds to multiplication by a term proportional to frequency, so it acts as a high-pass filter: edges (sharp transitions) are high-frequency content, and flat regions (near-zero derivative) are low-frequency content.

3.2 The Laplacian as an Ideal High-Pass Filter

Applying the theorem twice: $F\{\text{Laplacian}(f)\} = -(2\pi)^2(u^2+v^2)F(u,v)$, a radially symmetric filter that is zero at the origin and grows quadratically with frequency - the frequency-domain form of the Marr-Hildreth (Laplacian-of-Gaussian) edge detector.

3.3 Practical High-Pass Edge Filters and Low-Pass Region Filters

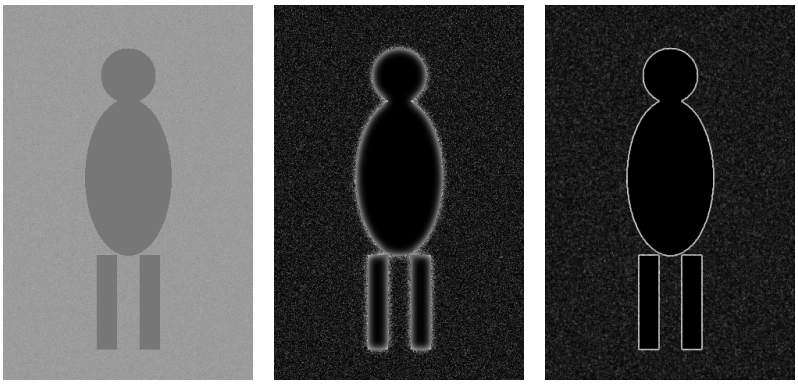
A Gaussian high-pass filter $H_{HP}(u,v) = 1 - \exp(-D(u,v)^2/2\sigma^2)$ isolates edges when multiplied with F(u,v); its complement, the Gaussian low-pass filter $H_{LP} = \exp(-D^2/2\sigma^2)$, smooths region interiors and is the frequency-domain justification for pre-blurring before region-based segmentation (thresholding, watershed): removing high-frequency noise/texture makes each region more internally uniform.

3.4 Band-Pass Filtering for Texture-Based Regions

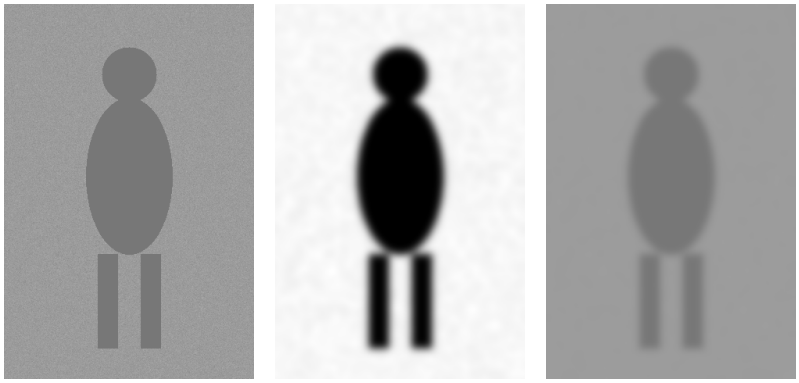
Textured regions concentrate energy in the power spectrum $|F(u,v)|^2$ at specific non-DC frequency bands. A band-pass filter (difference of two Gaussian low-pass filters at different cutoffs), or a bank of oriented Gabor filters, isolates regions sharing a characteristic spatial frequency/orientation even when their raw intensities overlap.

3.5 Experimental Validation

A Gaussian low-pass filter and its high-pass complement were applied in the frequency domain to a test silhouette image and compared against spatial-domain Gaussian blur and Sobel gradient magnitude respectively:



Original FFT high-pass Spatial Sobel

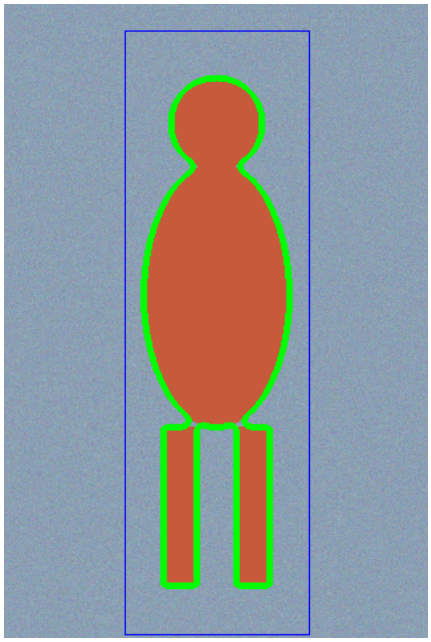


Original FFT low-pass Spatial Gaussian blur

The FFT high-pass result isolates the silhouette boundary (bright ring, dark interior/background), matching the spatial Sobel edge map's behavior. The FFT low-pass result smooths the image while preserving region structure, matching spatial Gaussian blur - confirming the theory in Sections 3.1-3.3.

4. Experimental Validation: Segmentation Results

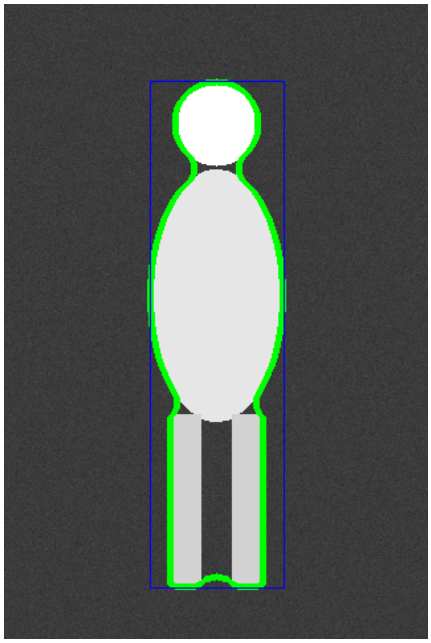
4.1 Part 1 — RGB Segmentation



Classical segmentation boundary (green) overlaid on the RGB test photo.

■ PLACEHOLDER: run compare_with_sam2.py on your own machine (SAM2 needs a GPU/checkpoint download not available in this sandboxed environment) and fill in RGB_IOU/RGB_DICE at the top of this script, then rebuild.

4.2 Part 2 — Thermal Segmentation



Classical segmentation boundary (green) overlaid on the thermal test image.

■ PLACEHOLDER: run compare_with_sam2.py on your real thermal photo, fill in THERMAL_IOU/THERMAL_DICE at the top of this script, then rebuild.

5. Conclusion

This report documents classical (non-ML) human boundary segmentation for RGB and thermal imagery, its comparison against SAM2 as an external deep-learning baseline, and the mathematical basis for frequency-domain edge detection and region segmentation, validated experimentally in Section 3.5.

6. Demonstration Video

See the submitted screen recording for a live demonstration of the web application segmenting both an RGB and a thermal image.